

Draft - expert review pending

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Topic 3.4 - Risk assessment and management

A real classroom situation

Back from the hazard-spotting walk (Topic 3.3), a pair of trainees argue. One has put "the broken window" at the top of the cards; the other insists on "the uncovered water tank". The first says, "A cut is a certainty; someone will lean on that frame this week." The second answers, "But the tank is a drowning; one child, once, is enough."

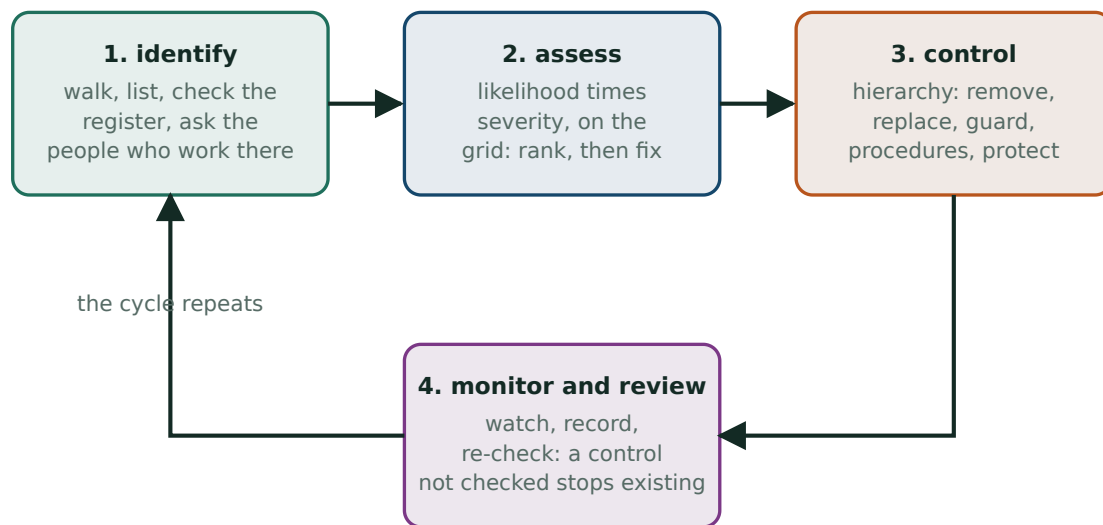
They have discovered, without being taught it, the two axes of risk: likelihood and severity. This topic gives their argument a name, a grid, and a process.

Explanation

Hazard identification, risk assessment and the risk management process

A hazard is the potential; risk is the chance and the seriousness of the harm actually occurring, in real conditions. The broken window and the open tank are both hazards; their risks differ in both directions. Risk is judged on two axes, how likely, and how severe, and the classic tool is a simple grid.

The risk management cycle



Done before someone is hurt, this is the whole method; the form nobody walked the building for is decoration.

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Step one: identify. Find the hazards before they find someone. The tools are ordinary: the walk-around inspection, the checklist, the incident register, and the most under-used one, asking the people who work there, who usually know. A Karachi site study found its hazards through exactly such community and site methods (Abbasi et al., 2022).

Hazard identification tools

Tool	What it finds	Its blind spot
Walk-around inspection	what is visible now: the broken window, the open tank, the wire	what only appears at night, in rush hour, or in the rain
Checklists	the known hazards, reliably: nothing on the list is missed	anything not on the list, including the new and unusual
Incident records	what has already hurt someone: patterns, repeat spots	near-misses nobody reported; the first occurrence
Asking the workers	what actually happens: the shortcut, the smell, the near-miss	only as good as the trust behind the asking

The most under-used tool is the last one: the people who work there usually know.
A Karachi site study found its hazards through exactly such community and site methods.

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Step two: assess. Score each hazard on likelihood and severity, low, medium, high is enough precision for a school. The broken window scores high likelihood, medium severity; the open tank scores lower likelihood, highest severity. Both land in the act-now corner, which is the point of the grid: it ranks, so the loudest hazard is not automatically the first one fixed.

Step three: control. Controls have a hierarchy, best to worst: remove the hazard entirely; replace it with something safer; guard or barrier between hazard and person; procedures and training; and last, personal protective equipment. PPE is last because it protects only one person, only while worn, only if correct. The Karachi study's finding, missing and unused protection, is exactly what a reliance on the bottom of the hierarchy looks like in practice.

Step four: monitor and review. A control that is not checked quietly stops existing: the guard is removed for convenience, the checklist stops being filled. The cycle repeats, which is why it is drawn as a loop.

A worked example, completed. The arguing pair take their two cards to the grid. The window: high likelihood, medium severity, act now; control, board it up (removal, the top of the hierarchy), cost, nothing. The tank: medium likelihood, highest severity, act now; control, a lockable cover (a barrier), cost, one visit to

the market. Both done the same afternoon. The rest of their cards, the generator exhaust, the unlit stair, the stress of exam week, are scored, ranked, and two go on the head teacher's desk with dates.

Common misconception. "Risk assessment is paperwork done after the danger is controlled." Read backwards, that is exactly wrong: the paperwork is the walk, the grid and the choice of control, done before someone is hurt. A form nobody walked the building for is not a risk assessment; it is a decoration.

Activity: Score and defend

Work in pairs. Take the five hazard cards from Topic 3.3's walk. Score each on the likelihood-severity grid, agree a control for each using the hierarchy, and prepare to defend one choice to another pair who will argue for a different order of fixing. Allow 25 minutes, then ten minutes of cross-examination.

Check your understanding

1. Define risk, and explain its two axes with the window and tank examples.
(*Understand*)
2. List the five levels of the hierarchy of controls from best to worst.
(*Remember*)
3. Why does personal protective equipment sit at the bottom of the hierarchy?
(*Analyze*)
4. Score these on the grid and justify the order you would fix them: a loose electric wire in a corridor (high likelihood, high severity), a wobbly ladder used once a year (low likelihood, high severity), exam-week stress (high likelihood, low severity). (*Apply*)

Summary

Risk is the chance and seriousness of harm, judged on likelihood and severity, and ranked on a grid so fixing follows evidence rather than noise. The risk management cycle is identify, assess, control, monitor, and it repeats. Controls follow a hierarchy, removal best and personal protection last, and the Karachi

site evidence shows what relying on the bottom of it produces. The walk, the grid and the hierarchy are ordinary tools; used before an injury, they are the whole of the method.

Self-assessment checklist

Sign in to sync your ticks across devices.

- ☐ I can define risk and distinguish it from hazard.
- ☐ I can score a hazard on the likelihood-severity grid.
- ☐ I can list the hierarchy of controls and say why PPE is last.
- ☐ I can run the four-step cycle on a real building and hand over a ranked list.

Try this at your practicum school

Turn your hazard walk into a mini risk assessment: five hazards, each scored on the grid, each with one control named and its level in the hierarchy. Hand the list to your cooperating teacher, and when you visit again, check which controls happened. Record the difference.

Summative task

Write 170 to 200 words on the following.

Write a one-page risk assessment brief for the school fair: the food stalls, the generator, the bouncy ride and the crowd. Identify at least four hazards across at least three classes, score them on the grid, and name a control for each with its place in the hierarchy. Finish with your monitoring plan for the day.

Mini-rubric

Criterion	What a strong response does
Cycle applied (Apply)	Identification, grid scoring and controls all present and matched
Hierarchy used (Understand)	Each control sits at a named level, not all at the bottom
Monitoring realistic (Analyze)	A check that would actually run on a fair day, with a person named
Clarity and register	Plain sentences for an organiser; within the word count

Further reading

- Abbasi, H. N., Khan, R., Nasir, I., & Ahmed, W. (2022). A case study of Tariq Road Underpass, Karachi: Environmental analysis. *International Journal of Economic and Environmental Geology*, 12(4).
<https://doi.org/10.46660/ijeeg.Vol12.Iss4.2021.651>
- World Health Organization. *Occupational health* (health topic):
<https://www.who.int/health-topics/occupational-health>
- Harrington, J. M., & Gill, F. S. (Eds.). *Occupational Health*. Blackwell. (guide-recommended book, cited by reference only)

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